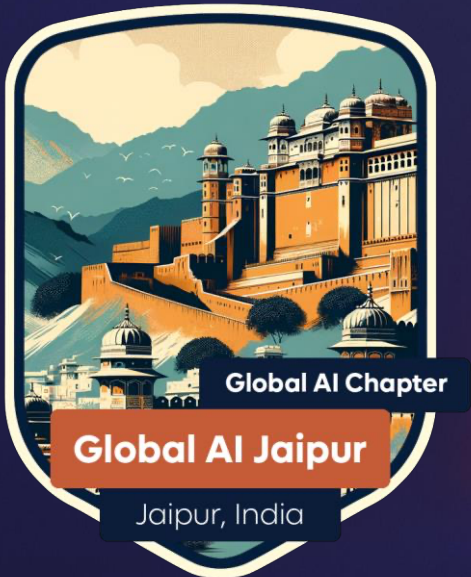
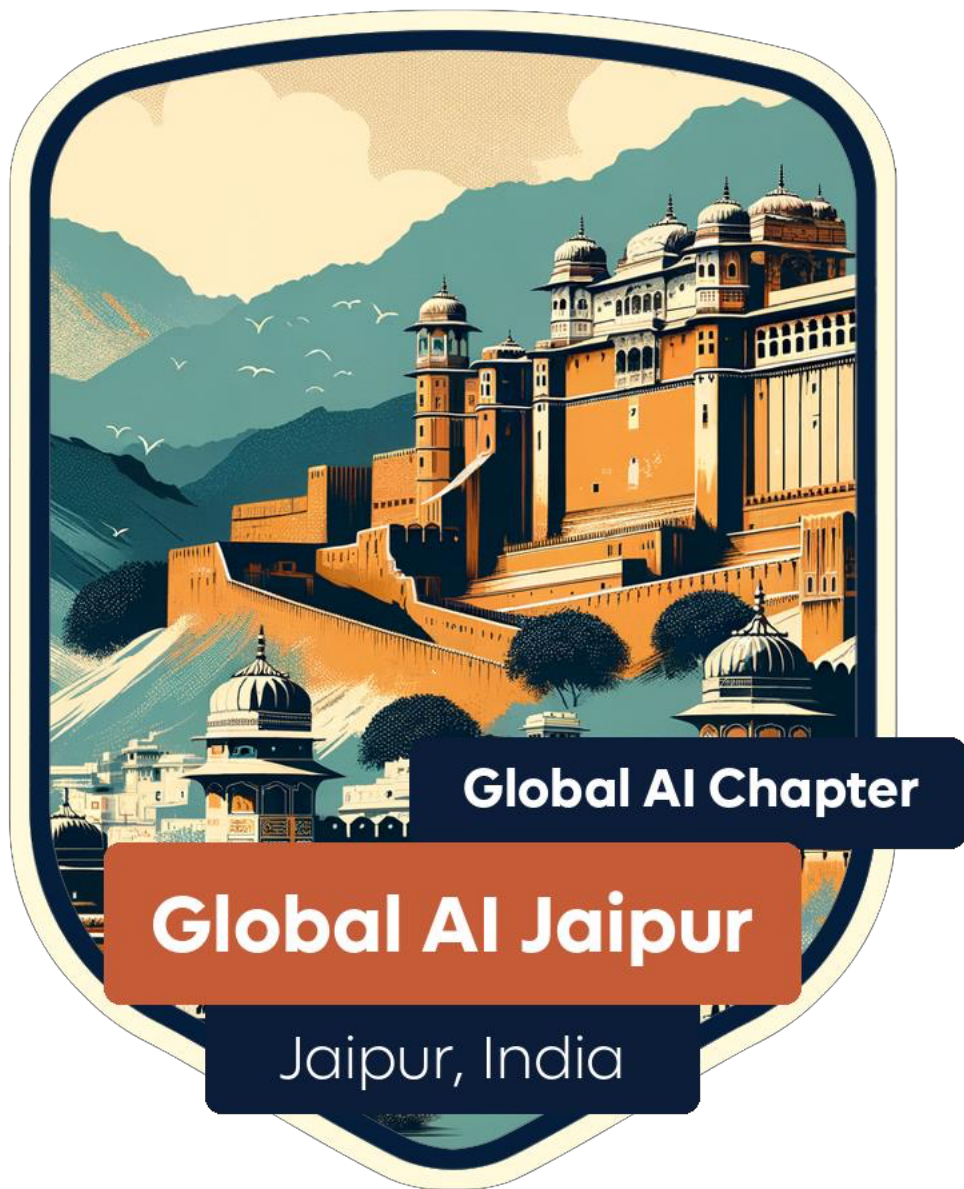


Fundamentals OF LLM & RAG



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- LLMs
- How to use LLMs
- Some Interesting Tips
- Q&A

LLMs

Large Language Models

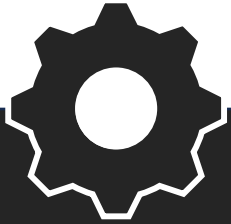
LLM is a deep learning model trained on massive text datasets. Designed to understand, generate, and interact with human language.

Based on Transformer architecture. Learns patterns, context, and meaning across billions of words.

Can perform tasks like:

- Text generation
- Translation
- Summarization
- Question answering

LLMs Timeline



Pre-Transformer

1990 - 2017

RNN, LSTM, NMT, ELMo model

Transformer

2017 - 2018

The rise of Transformers - No specific LLMs Yet...

Post-Transformer

2018 - Present

GPT-1, BERT – GPT-2, XLNet – GPT3, T5 - BLOOM, Codex, GPT-4

Attention Is All You Need

Attention Is All You Need

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Abstract

The dominant sequence transduction models are based on complex recurrent or convolutional neural networks that include an encoder and a decoder. The best performing models also connect the encoder and decoder through an attention mechanism. We propose a new simple network architecture, the Transformer, based solely on attention mechanisms, dispensing with recurrence and convolutions entirely. Experiments on two machine translation tasks show these models to be superior in quality while being more parallelizable and requiring significantly less time to train. Our model achieves 28.4 BLEU on the WMT 2014 English-to-German translation task, improving over the existing best results, including ensembles, by over 2 BLEU. On the WMT 2014 English-to-French translation task, our model establishes a new single-model state-of-the-art BLEU score of 41.0 after training for 3.5 days on eight GPUs, a small fraction of the training costs of the best models from the literature.

Attention Is All You Need

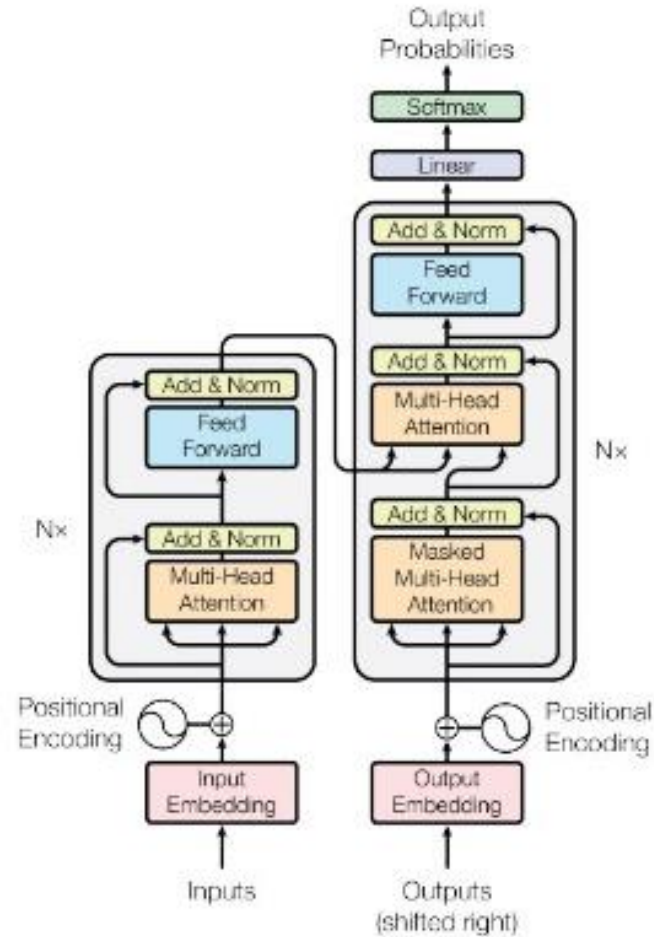


Figure 1: The Transformer - model architecture.

wise fully connected feed-forward network. We employ a residual connection [10] around each of the two sub-layers, followed by layer normalization [11]. That is, the output of each sub-layer is $\text{LayerNorm}(x + \text{Sublayer}(x))$ where $\text{Sublayer}(x)$ is the function implemented by the sub-layer

ChatGPT
was trained
on over **300 billion**
words!



Why Are LLMs Transformative?



Speed

Process and analyze
vast amounts of text
in seconds.

Why Are LLMs Transformative?



Scale

Operate in
multiple languages
and domains.

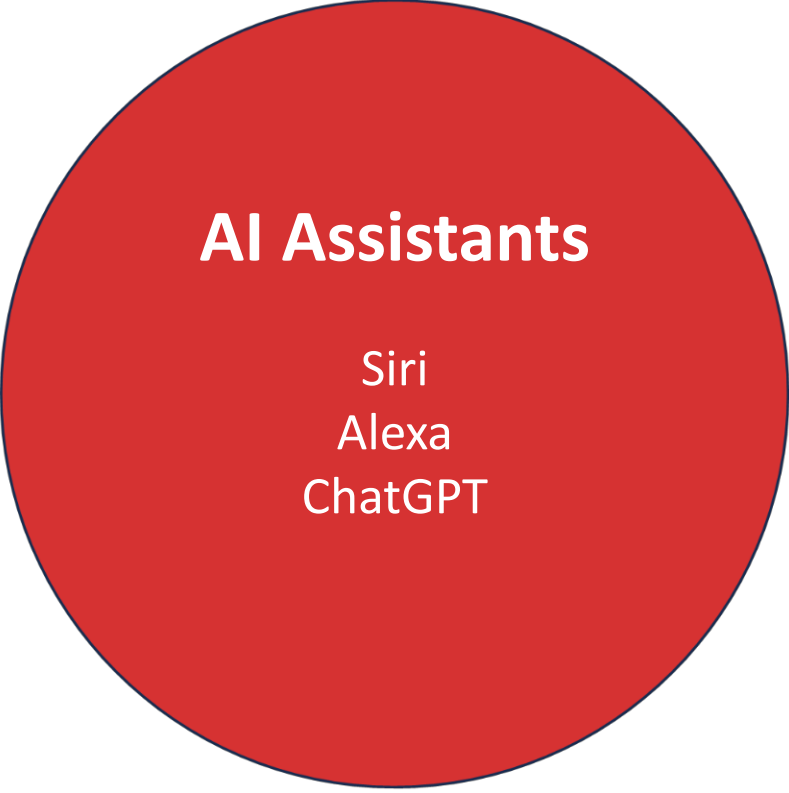
Why Are LLMs Transformative?



Intelligence

Generate creative,
coherent,
and useful content.

LLMs in Everyday Life



AI Assistants

Siri
Alexa
ChatGPT

LLMs in Everyday Life



Content Creation

Articles
Code
even poetry

LLMs in Everyday Life



Customer Support

Smarter chatbots
and faster solutions

We use language models every day



what is the | 

- what is the **weather**
- what is the **meaning of life**
- what is the **dark web**
- what is the **xfl**
- what is the **doomsday clock**
- what is the **weather today**
- what is the **keto diet**
- what is the **american dream**
- what is the **speed of light**
- what is the **bill of rights**



How it all started?

Next Word Prediction

INPUT



OUTPUT

How it all started?

Next Word Prediction

The Car is ----



How it all started?

Next Word Prediction

The Car is ----



Red

How it all started?

Next Word Prediction

The Car is red



How it all started?

Next Word Prediction

Tell a story.



How it all started?

Next Word Prediction

Tell a story.



Once

How it all started?

Next Word Prediction

Tell a story. Once



How it all started?

Next Word Prediction

Tell a story. Once



upon

How it all started?

Next Word Prediction

Tell a story. Once upon



How it all started?

Next Word Prediction

Tell a story. Once upon



a time

How it all started?

Next Word Prediction

Tell a story. Once upon a time

.....



A little deeper

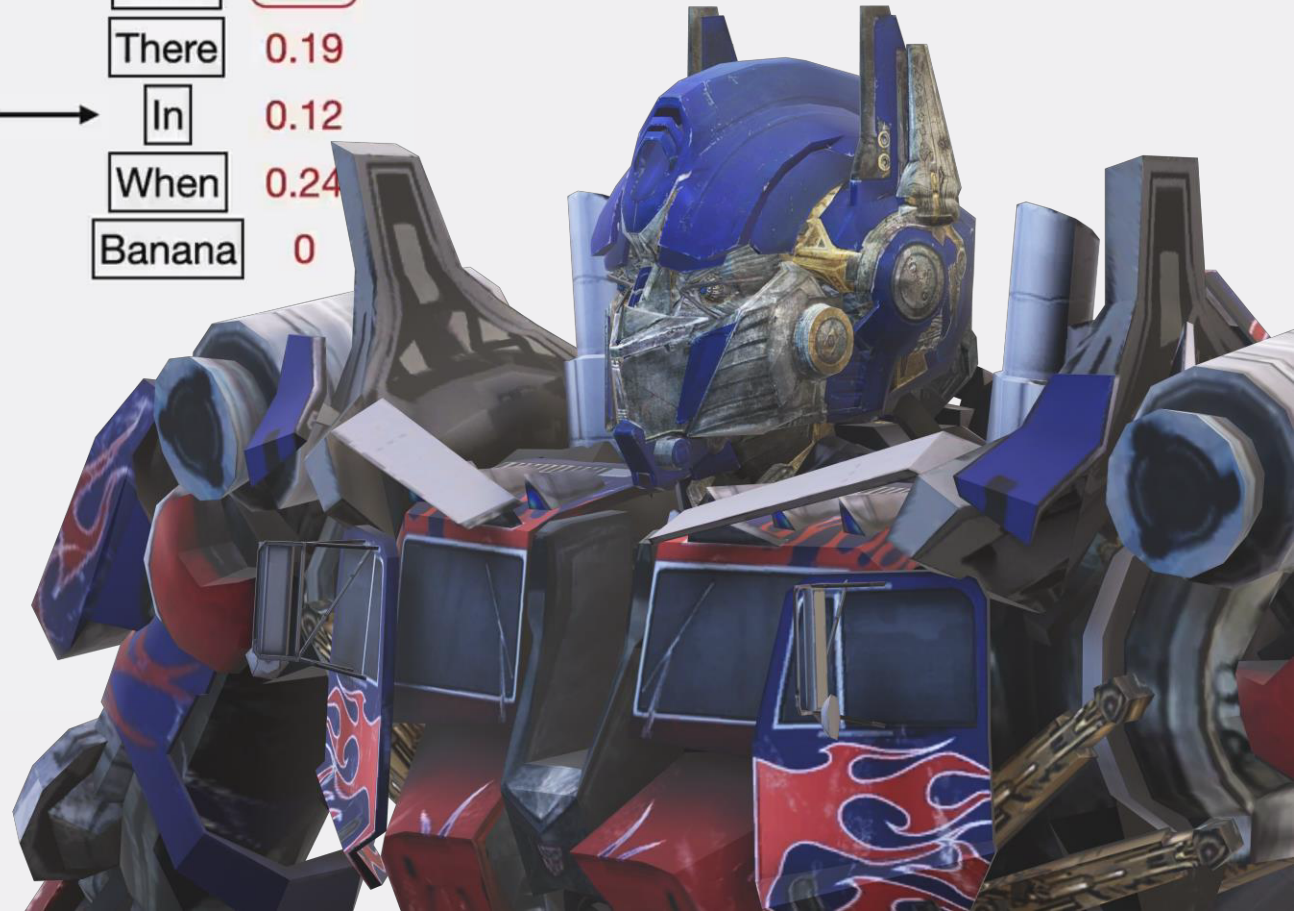
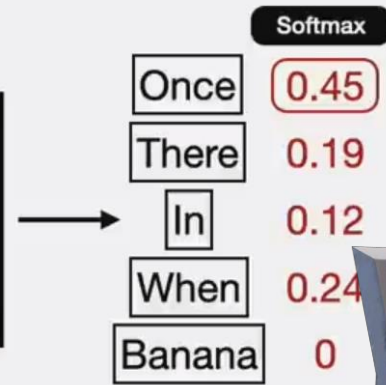
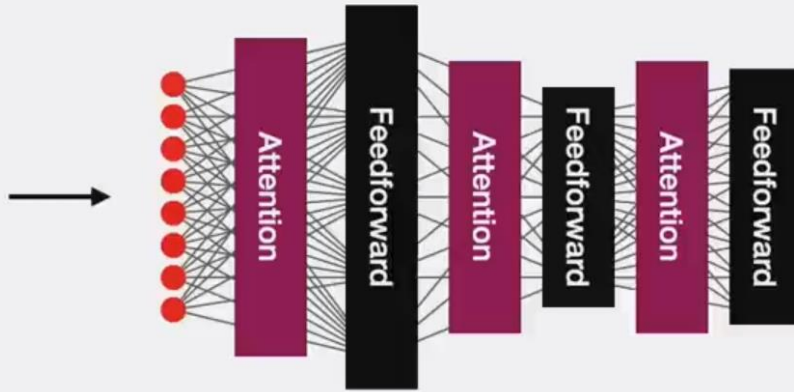
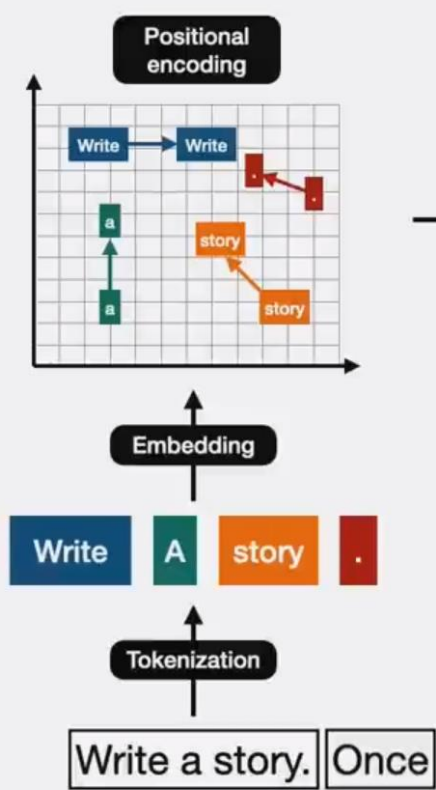
How it all started?

Write a story.



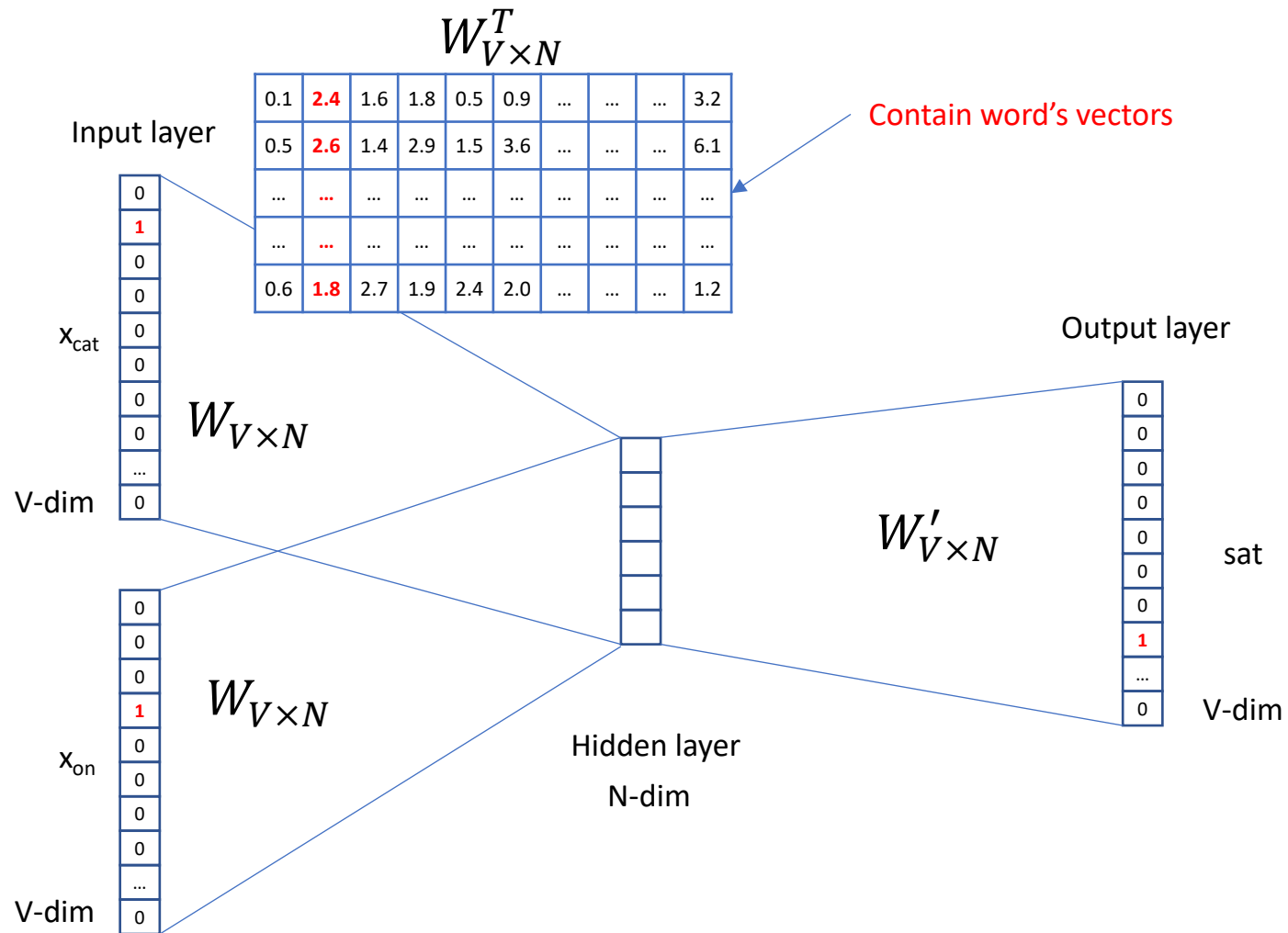
A little deeper

How it all started?



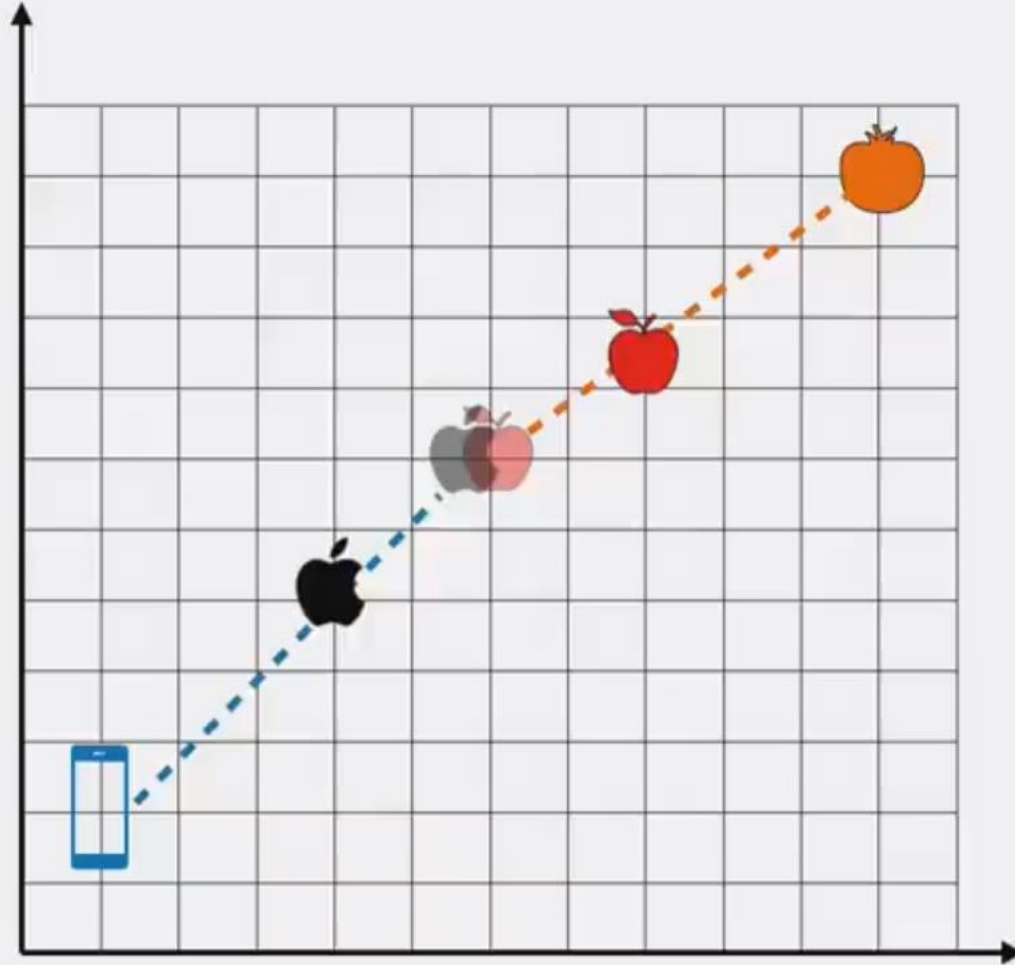
How Do Models Work?

- At their most basic models are statistical pattern-recognition and prediction systems
- They output the next likely word (“token”) in a sentence (“sequence”)
 - token: unit of text e.g. word, character. 1 word ~ 0.75 token
 - sequence: context - section (“window”) of text e.g. sentence, paragraph, book
 - input into chatGPT is 4096 tokens; Claude 2 is 100K tokens
- The likelihood of the next word appearing is determined by
 - the context in which the words are seen in a larger body of text (“corpus”) and
 - the input to the chat



We can consider either W or W' as the word's representation. Or even take the average.

Words pulling words



please buy an **apple** and an **orange**

apple unveiled the new **phone**

n-gram Language Models

- A **basic language model** used to predict the next word in a sequence.
- It looks at the **previous (N-1) words** to predict the next one.
- **N** refers to how many words are considered:
 - **Unigram** (N=1): Only the current word.
 - **Bigram** (N=2): Current + previous word.
 - **Trigram** (N=3): Current + two previous words.
- Works based on the **probability of word sequences**.

1-gram

Hello, how are happy
1 word

Unigram probability: $P(w) = \frac{\text{count}(w)}{m} = \frac{C(w)}{m}$

Data

... are **you** ...

... are **sad** ...

... are?...

... are **happy** ...

... are **ready** ...

... are **happy** ...

... are **free** ...

3-gram

Hello, how are _____

3 words

$$P(w_3|w_1 w_2) = \frac{C(w_1, w_2, w_3)}{\sum_w C(w_1, w_2, w)} = \frac{C(w_1, w_2, w_3)}{C(w_1, w_2)}$$

Data

Hello, how are **you**?

Hello, how are **things** going?

Hello, how are **things** today?

Hello, how are **the** kids?

Hello, how are **the** others?

Hello, how are **they** doing?

Hello, how are **things** happening?

10-gram

Hello, today I would like
to inform you that ____

10 words

$$P(A | B) = \frac{P(A,B)}{P(B)}$$

$$P(w_i | w_1, w_2, \dots, w_{i-1}) = \frac{C(w_1, w_2, \dots, w_{i-1}, w_i)}{C(w_1, w_2, \dots, w_{i-1})}$$

Data

Hello, today I would like to inform
you that **the**

Hello, today I would like to inform
you that **the**

Transformers

A deep learning model architecture for sequence data (text, images, etc.).

Uses **Self-Attention** to focus on important parts of the input.

Learns **relationships** between words or parts of the input, regardless of their position.

Used in:

- Chatbots (like ChatGPT)

- Translation (Google Translate)

- Image recognition (Vision Transformers)

- Music, video, and drug discovery.

Why did the transformer make such a big difference for language modeling?

It allowed for faster learning of more model parameters on more data (by allowing parallel computation on GPUs!)

Transformers

Key Components

Input Embeddings: Convert text to numerical vectors.

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Positional Encoding: Capture the order of words.

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Transformers

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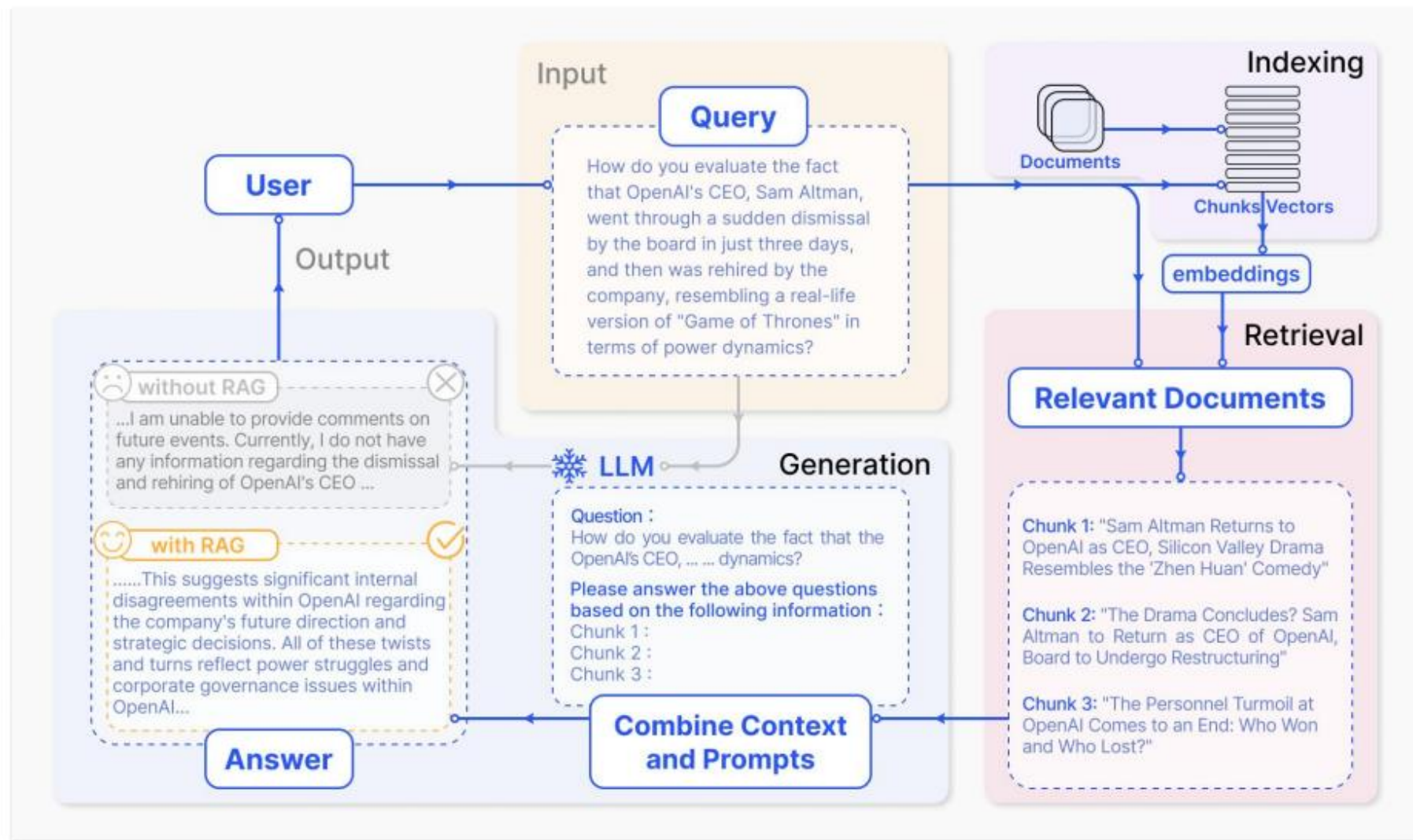
Result: Complex understanding of long-range dependencies.

► Retrieval-Augmented Generation (RAG)

When answering questions or generating text, it first **retrieves relevant information** from a large number of documents, and then LLMs generates answers based on this information.

By attaching a **external knowledge base**, there is no need to retrain the entire large model for each specific task.

The RAG model is especially suitable for **knowledge-intensive** tasks.



A typical case of RAG

► RAG Applications

Scenarios where RAG is applicable:

- Long-tail distribution of data
- Frequent knowledge updates
- Answers requiring verification and traceability
- Specialized domain knowledge
- Data privacy preservation

Q&A

RETRO (Borgeaud et al, 2021)
REALM (Gu et al, 2020)
ATLAS (Izacard et al, 2023)

Fact Checking

RAG (Lewis et al, 2020)
ATLAS (Izacard et al, 2022)
Evi. Generator (Asai et al, 2022a)

Dialog

BlenderBot3 (Shuster et al, 2022)
Internet-augmented generation (Komeili et al., 2022)

Summary

FLARE (Jiang et al, 2023)

Machine Translation

kNN-MT (Khandelwal et al., 2020)
TRIME-MT (Zhong et al., 2022)

Code Generation

DocPrompting (Zhou et al., 2023)
Natural Prover Welleck et al., 2022)

Natural Language Inference

kNN-Prompt (Shi et al., 2022)
NPM (Min et al., 2023)

Sentiment analysis

kNN-Prompt (Shi et al., 2022)
NPM (Min et al., 2023)

Commonsense reasoning

Raco (Yu et al, 2022)

Challenges in LLM



Lack of True Understanding



LLMs don't "understand" text the way humans do.
They rely on **statistical patterns** in data rather than reasoning or comprehension.

Challenges in LLM

[Dependence on Input Phrasing]

LLMs are sensitive to the wording of **prompts**, which can drastically affect their output.

Challenges in LLM

Hallucinations and Fabrication

LLMs can "**hallucinate**" facts,
generating entirely false or misleading information **with confidence**.



Challenges in LLM

Resource Intensive

LLMs require **vast computational resources** for training and inference.

Challenges in LLM

[Bias in Outputs]

LLMs **inherit biases** present in their **training data**, which can lead to unfair or **inappropriate outputs**.

Challenges in LLM



Limited Domain-Specific Expertise



While LLMs are **generalists**,
they may **struggle** with **highly specialized** tasks without fine-tuning.

Challenges in LLM



Ethical and Security Concerns



LLMs can be **misused** to **generate harmful content** like misinformation, phishing attacks, or deepfakes.

LLMs we can use...

GPT-3

GPT-4

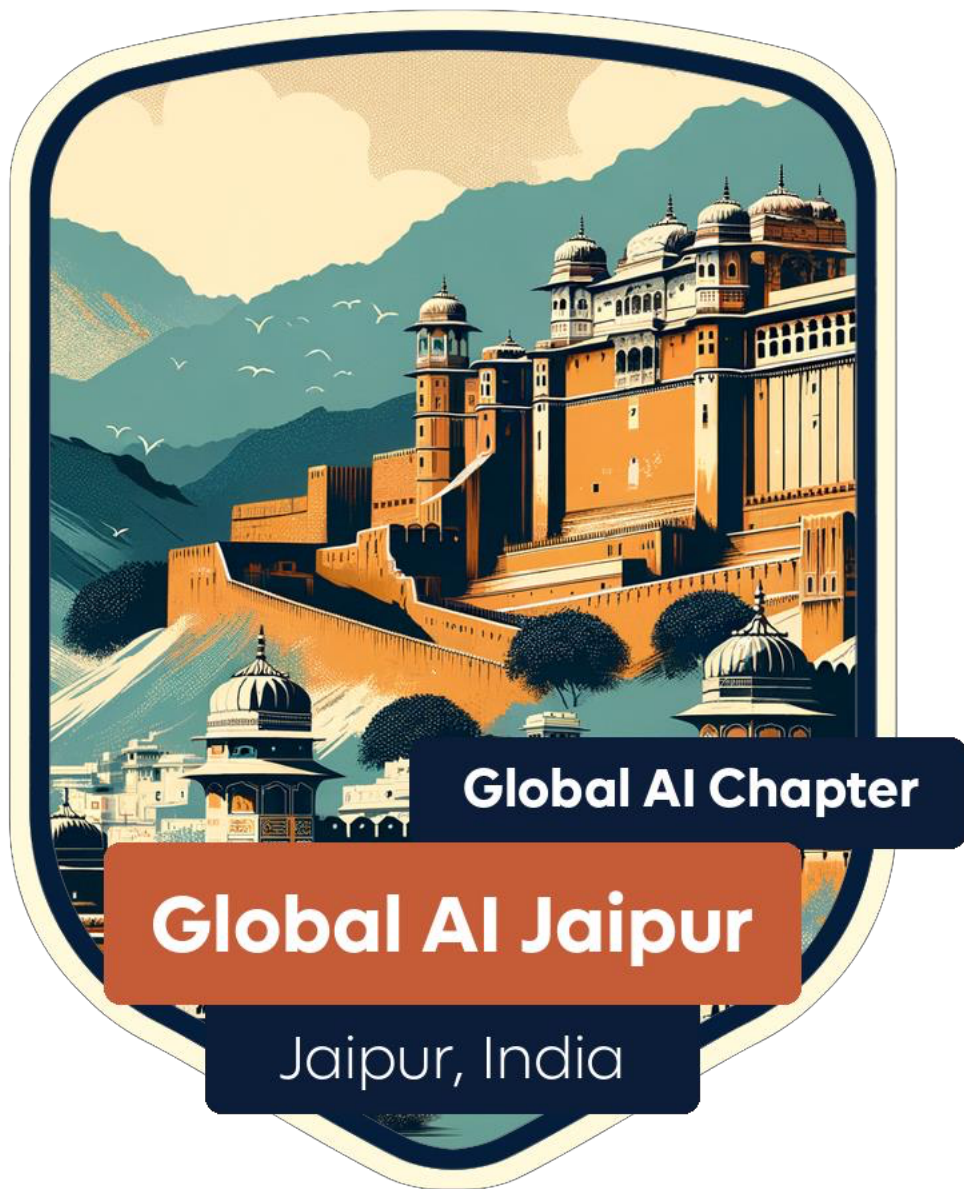
Codex-OpenAI

Claude

Deep Seek

Gemini

Llama



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